

EDUSAMBAM ASSESSMENT SERIES

Practice Test: Angle Properties, Polygons & Circle Theorems

PRINTABLE PRACTICE WORKSHEET

EDUSAMBAM, working on building your Mathematics understanding. Visit edusambam.com for the Auto-graded Self-assessment Experience.

YOUR PHYSICAL & ONLINE SELF-ASSESSMENT ROADMAP

PHYSICAL

1
Download any
practice test

2
Solve within
assigned time

3
Submit it to your
Teacher/Examiner

4
Get assessed
and improve

ONLINE

1
Visit
edusambam.com

2
Open Assessment
Arena

3
Solve online
papers

4
Get auto-assessed
and improve

Candidate Name:		Roll No.:	
Class / Grade:		Date:	
School / Institution:		Test Centre:	

Subject	Mathematics (Maths)	Topic	Practice Test: Angle Properties, Polygons & Circle Theorems	Total Marks	100
Total Questions	45	Time Allowed	1 Hour (60 minutes)	Version	A

GENERAL INSTRUCTIONS

1. Read each question carefully before answering.
2. For Section A and Section B, mark one answer only.
3. For Sections C and E, write your answer clearly in the space provided.
4. For Section D, write the correct letter from Column B against each item in Column A.
5. Do not use correction fluid. Keep all answers clear and legible.
6. This worksheet is prepared from the EDUSAMBAM practice assessment supplied for this paper.

PAPER PATTERN

Section	Questions	Marks Each	Total
Section A — Multiple Choice	15	2	30
Section B — True / False	10	2	20
Section C — Fill in the Blank	10	2	20
Section D — Column Matching	5 pairs	2	10
Section E — Short Answer	5	4	20

Total	45	—	100
-------	----	---	-----

Section A — Multiple Choice

30 marks

1. Angles on a straight line always sum to:

Basic

- A. 90° B. 360°
C. 180° D. 270°

Answer: ☐ A ☐ B ☐ C ☐ D

2. Where two straight lines cross, vertically opposite angles are always:

Basic

- A. Equal B. Supplementary
C. Complementary D. Always 90°

Answer: ☐ A ☐ B ☐ C ☐ D

3. A transversal crosses two parallel lines. The alternate angles formed are:

Intermediate

- A. Supplementary B. Always 90°
C. Unrelated D. Equal

Answer: ☐ A ☐ B ☐ C ☐ D

4. A triangle's two interior opposite angles are 50° and 65° . Its exterior angle at the third vertex is:

Intermediate

- A. 65° B. 115°
C. 180° D. 50°

Answer: ☐ A ☐ B ☐ C ☐ D

5. The interior angle sum of a hexagon (6 sides) is:

Intermediate

- A. 540° B. 900°
C. 360° D. 720°

Answer: ☐ A ☐ B ☐ C ☐ D

6. The exterior angles of any convex polygon always sum to:

Intermediate

- A. 180° B. 360°
C. $(n - 2) \times 180^\circ$ D. It depends on the number of sides

Answer: ☐ A ☐ B ☐ C ☐ D

7. A regular polygon has an exterior angle of 45° . How many sides does it have?

Advanced

- A. 8 B. 6
C. 9 D. 5

Answer: ☐ A ☐ B ☐ C ☐ D

8. AB is a diameter of a circle, and C is a point on the circumference. Angle ACB equals:

Basic

- A. 45° B. 180°
C. 90° D. 60°

Answer: ☐ A ☐ B ☐ C ☐ D

9. The angle at the centre of a circle, compared to the angle at the circumference on the same arc, is:

Intermediate

- A. Half as large B. Twice as large
C. Equal D. Three times as large

Answer: ☐ A ☐ B ☐ C ☐ D

10. Angles in the same segment, subtending the same arc, are:

Intermediate

- A. Supplementary B. Complementary
C. Unrelated D. Equal

Answer: ☐ A ☐ B ☐ C ☐ D

11. In a cyclic quadrilateral, opposite angles always:

Intermediate

- A. Are equal
B. Sum to 360°
C. Sum to 180°
D. Sum to 90°

Answer: ☐ A ☐ B ☐ C ☐ D

12. A tangent meets the radius drawn to the point of contact at:

Basic

- A. 90°
B. 45°
C. 60°
D. It varies

Answer: ☐ A ☐ B ☐ C ☐ D

13. Two tangents drawn from the same external point to a circle are:

Intermediate

- A. Equal in length
B. Parallel
C. Perpendicular to each other
D. Never equal

Answer: ☐ A ☐ B ☐ C ☐ D

14. The alternate segment theorem relates the tangent-chord angle to:

Advanced

- A. The angle at the centre
B. The radius length
C. The angle in the alternate segment
D. The diameter

Answer: ☐ A ☐ B ☐ C ☐ D

15. The region between a chord and an arc is called a:

Basic

- A. Sector
B. Segment
C. Tangent
D. Radius

Answer: ☐ A ☐ B ☐ C ☐ D

Section B — True / False

20 marks

16. Angles around a single point always sum to 360° .

Basic

☐ True ☐ False

17. In an isosceles triangle, the two base angles opposite the equal sides are equal.

Intermediate

☐ True ☐ False

18. A quadrilateral's interior angles always sum to 180° .

Basic

☐ True ☐ False

19. Co-interior (allied) angles between two parallel lines sum to 180° .

Intermediate

☐ True ☐ False

20. A regular polygon with n sides has n lines of symmetry.

Intermediate

☐ True ☐ False

21. AB is a diameter of a circle. Any angle drawn from a point on the circumference to A and B is always 90° .

Basic

☐ True ☐ False

22. The angle at the centre of a circle is always half the angle at the circumference on the same arc.

Intermediate

☐ True ☐ False

23. In a cyclic quadrilateral, opposite angles are always equal to each other.

Intermediate

☐ True ☐ False

24. A tangent to a circle touches it at exactly one point.

Basic

☐ True ☐ False

25. The alternate segment theorem applies at the exact point where the tangent touches the circle.

Advanced

☐ True ☐ False

Section C — Fill in the Blank

20 marks

26. Angles on a straight line always sum to _____ degrees.

Answer:

27. When two lines cross, the equal angles directly opposite each other are called _____ angles.

Answer:

28. The interior angle sum of a polygon with n sides is $(n - 2) \times$ _____.

Answer:

29. A polygon with all sides and all angles equal is called a _____ polygon.

Answer:

30. The exterior angles of any convex polygon always sum to _____ degrees.

Answer:

31. The longest chord of a circle, passing through the centre, is called the _____.

Answer:

32. An angle formed in a semicircle using the diameter is always a _____ angle.

Answer:

33. A quadrilateral with all four vertices on a circle is called a _____ quadrilateral.

Answer:

34. A tangent meets the radius drawn to the point of contact at _____ degrees.

Answer:

35. Two tangents drawn from the same external point to a circle are always _____ in length.

Answer:

Section D — Column Matching

10 marks

36–40. Match each term to its correct definition.

Column A	Ans.	Column B
Exterior angle theorem	_____	A. A straight line that touches a circle at exactly one point
Cyclic quadrilateral	_____	B. Equals the sum of the two interior opposite angles
Tangent	_____	C. The tangent-chord angle equals the angle in the alternate segment
Regular polygon	_____	D. A four-sided shape with every vertex on a circle
Alternate segment theorem	_____	E. A polygon with all sides and all angles equal

Section E — Short Answer

20 marks

41. Explain why the exterior angle of a triangle equals the sum of the two interior opposite angles.

42. Describe how to find the number of sides of a regular polygon if you know its exterior angle.

43. Explain what makes a triangle isosceles, and how this affects its angles.

44. Describe the relationship between the angle at the centre and the angle at the circumference of a circle, for the same arc.

45. Explain why the angle in a semicircle is always 90° .

Candidate Signature: _____

Invigilator Signature: _____

FOR OFFICE USE ONLY

Marks Obtained: _____ / 100 Checked By: _____

Result / Grade: _____ Date Checked: _____

Practice Test: Angle Properties, Polygons & Circle Theorems — Answer Key

Teacher reference: equivalent grammatically correct wording is acceptable for open-ended items when the meaning is preserved.

Section A — Multiple Choice

- | | |
|------|-------|
| 1. C | |
| 2. A | 9. B |
| 3. D | 10. D |
| 4. B | 11. C |
| 5. D | 12. A |
| 6. B | 13. A |
| 7. A | 14. C |
| 8. C | 15. B |

Section B — True / False

- | | |
|-----------|-----------|
| 16. True | 21. True |
| 17. True | 22. False |
| 18. False | 23. False |
| 19. True | 24. True |
| 20. True | 25. True |

Section C — Fill in the Blank

- | | |
|-------------------------|--------------|
| 26. 180 | 31. diameter |
| 27. vertically opposite | 32. right |
| 28. 180 | 33. cyclic |
| 29. regular | 34. 90 |
| 30. 360 | 35. equal |

Section D — Column Matching

- | | |
|--------------------------------|-----------------------------------|
| 36. Exterior angle theorem — B | |
| 37. Cyclic quadrilateral — D | 39. Regular polygon — E |
| 38. Tangent — A | 40. Alternate segment theorem — C |

Section E — Short Answer (Model Answer Keywords)

41. third angle; 180; straight line; sum
42. 360; divide; number of sides
43. two equal sides; base angles; equal
44. twice; double; centre
45. diameter; semicircle; 90; right angle